Planar Turán Numbers of Triangular Blocks

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Joint Math Meeting 2026

BSM

BUDAPEST SEMESTERS IN MATHEMATICS

Planar Turán Number

Definition

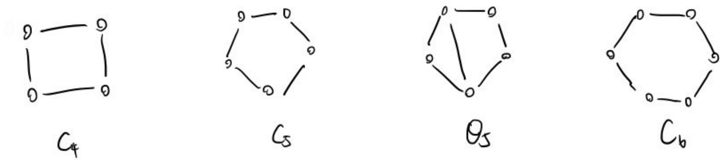
The planar Turán number of a graph H, denoted $\mathrm{ex}_{\mathrm{p}}(H, n)$, is defined to be the maximal number of edges of a planar graph on n vertices that doesn't contain H as a subgraph.

Note that $\mathrm{ex}_{\mathrm{p}}(H, n) \leqslant \mathrm{ex}_{\mathrm{p}}(-, n) = 3n - 6$. "$=$" holds if there is a triangulation on n vertices avoiding H.

Known Results

Theorem

1. $\mathrm{ex_p}(C_4, n) = \frac{15(n-2)}{7}$;
2. $\mathrm{ex_p}(C_5, n) = \frac{12n-33}{5}$;
3. $\mathrm{ex_p}(\theta_5, n) = \frac{5(n-2)}{2}$;
4. $\mathrm{ex_p}(C_6, n) = \frac{18(n-2)}{7}$.

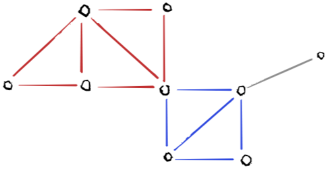


Triangular Blocks

Definition

A triangular block is a subgraph of a planar graph G obtained by the following algorithm:

1. Start with a list L containing one edge in G;
2. add edges that share a triangular face with some edge in L;
3. repeat until no more edges can be added; L spans the triangular block.



(an edge partition)

Triangular Blocks

Each edge contributes to multiple faces.

$$\mathrm{fc}(B) = \sum_{e \in E(B)} \left(\frac{1}{|f_1(e)|} + \frac{1}{|f_2(e)|} \right) \quad \text{is the face contribution}$$

$$\mathrm{ec}(B) = \#\ \text{edges in } B \quad \text{is the edge contribution}$$

Contribution Method

Observe that

$$\#\text{faces in } G = \sum_{\text{blocks } B \text{ in } G} \mathrm{fc}(B)$$

$$\#\text{edges in } G = \sum_{\text{blocks } B \text{ in } G} \mathrm{ec}(B)$$

Bounding the contributions lets us bound the number of edges!
(in terms of the number of faces)

Contribution Method

Sketch of Proof for $\mathrm{ex}(C_4, n) \leqslant \frac{15}{7}(n-2)$

The only triangular blocks avoiding C_4 are triangles and edges. By Euler, $n-2 = e-f$, so it suffices to show that each block satisfies

$$e \leqslant \frac{15}{7}(e-f) \quad \text{which simplifies to} \quad f \leqslant \frac{8}{15}e.$$

If G avoids C_4, then

- For B an edge in G, $\mathrm{fc}(B) \leqslant \frac{1}{2}$ and $\mathrm{ec}(B) = 1$.
- For B a triangle in G, $\mathrm{fc}(B) \leqslant \frac{8}{5}$ and $\mathrm{ec}(B) = 3$.

In both cases, $\mathrm{fc}(B) \leqslant \frac{8}{15}\mathrm{ec}(B)$, completing the proof.

New Perspective

What if we treat triangular blocks as the object of study?

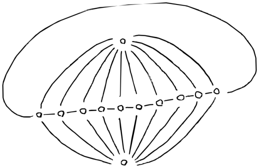
Question: *Can we characterize the planar Turán number of all triangular blocks?*

Result I

Theorem

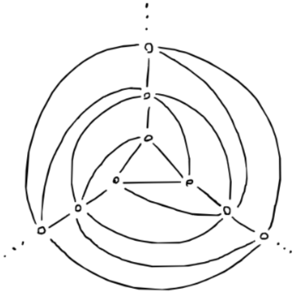
Let T be a triangular block with planar Turán Number $\mathrm{ex}_{\mathrm{p}}(T, n) < 3n - 6$ for all n sufficiently large. Then, T is a subgraph of the following.

$\Sigma_{12} =$



Sketch of Proof

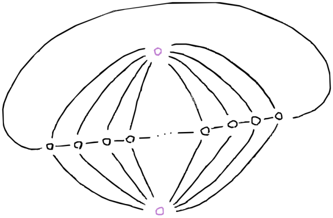
A triangulation with uniform degree 6

$\Delta_i =$ 

Sketch of Proof

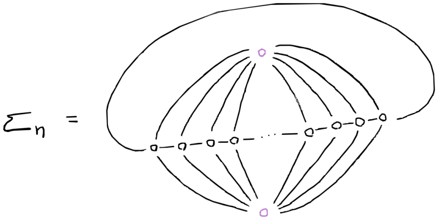
A ladder

$\Sigma_n =$



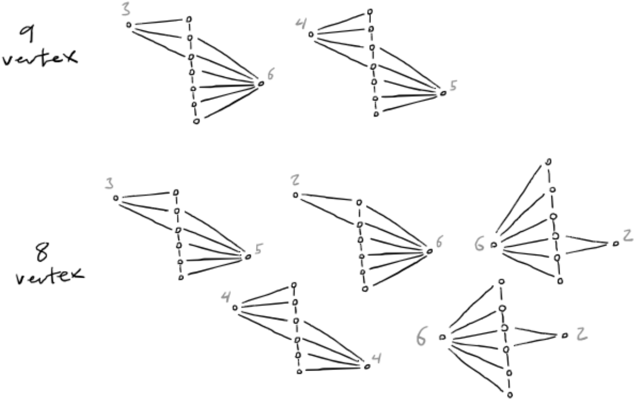
Sketch of Proof

A ladder

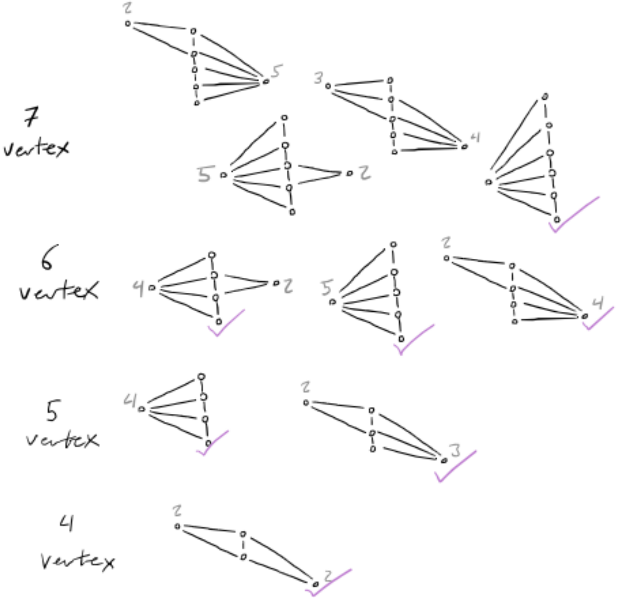


Any triangular block $H \subseteq \Sigma_n$ with > 10 vertices has a vertex of degree > 6

Triangular Blocks

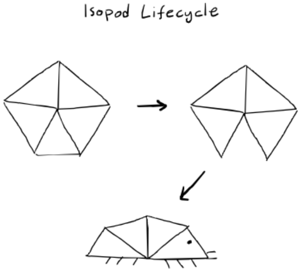


Triangular Blocks



Triangular Blocks

Isopods!



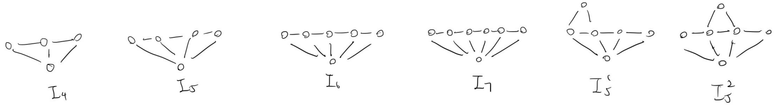
Result II

Theorem

For n sufficiently large, we have

1. $\mathrm{ex_p}(I_4, n) \leqslant \frac{12}{5}(n-2)$;
2. $\mathrm{ex_p}(I_5, n) \leqslant \frac{8}{3}(n-2)$;
3. $\mathrm{ex_p}(I_5^1, n) \leqslant \frac{36}{13}(n-2)$;
4. $\mathrm{ex_p}(I_5^2, n) \leqslant \frac{36}{13}(n-2)$;
5. $\mathrm{ex_p}(I_6, n) \leqslant \frac{48}{17}(n-2)$;
6. $\mathrm{ex_p}(I_7, n) \leqslant \frac{120}{41}(n-2)$.

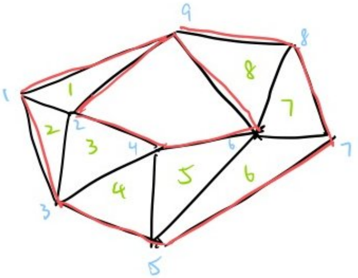
All of the above bounds have equality for infinitely many n.



Sketch of Proof for I_d

Given a triangular block with v vertices, e edges, f interior faces, k exterior faces, and s edges on the boundary.

$$e = 3v - s + 3(k-2);$$
$$f = 2v - 4 + 2k - s.$$



$v = 9$
$e = 17$
$f = 8$
$k = 2$
$s = 10$

Sketch of Proof for I_d

Lemma

$\mathrm{ex_p}(G, n) \leqslant \sup_{G \nsubseteq B} \frac{\mathrm{ec}(B)}{\mathrm{ec}(B)-\mathrm{fc}(B)}(n-2) \leqslant \sup_{G \nsubseteq B} \frac{3(v+k-2)-s}{(v+k-2)-\frac{s}{4}}$.

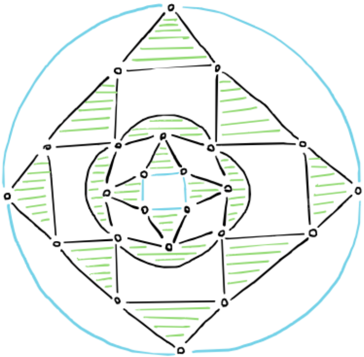
Lemma

If B avoids I_d, $2e = \sum_{v \in V} \deg(v) \leqslant (d-2)(k+v-1)$.

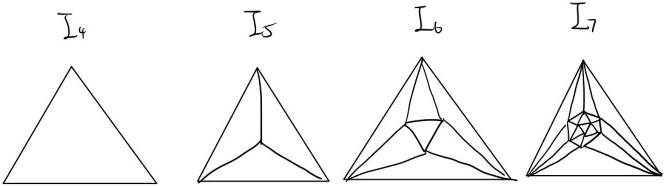
For $I_4 \nsubseteq B$, $\frac{e(B)}{e(B)-\mathrm{fc}(B)} \leqslant \frac{8v}{4v-2}$. When $v \geqslant 3$, $\frac{8v}{4v-2} \leqslant \frac{12}{5}$.

Construction

Skeleton



Construction



Future Work

Conjecture

All extremal constructions of triangular blocks uses the same skeleton.

Additional directions

- Fractional partitions;
- Higher genus graphs!

Thank you!